



NATIONAL INSTITUTE OF BUSINESS MANAGEMENT
DIPLOMA IN COMPUTER SOFTWARE ENGINEERING 18.1F

COMPUTER TECHNOLOGY – DSE-2-4-05

1st October 2018

9.00 am – 12.00 noon

Answer **five** questions.

Electronic calculators are allowed.

Time: THREE hours.

PART A-DIGITAL ELECTRONICS AND DIGITAL ELECTRONICS

Question 1

- a. Apply De Morgan's Theorems to the following expressions and **simplify**:

i. $Z = \overline{\overline{A + AB + ABC}(ABC + AC)}$

ii. $Z = \overline{(\overline{A.B.C} + C) + (A + B) + (\overline{A} + \overline{BC})}$

- b. $Z = F(A, B, C, D)$

$Z = 1$ for the minterms (2, 3, 8, 9, 12, 14)

$Z = \text{don't care}$ for the minterms (6, 7, 13)

$Z = 0$ for the remaining minterms

Simplify Z using

- I. K Map Method
- II. Tabular Method.

(20 Marks)

Question 2

- a) A car alarm system has four sensors A, B, C, and D sensors. A and B are connected to the two front doors while sensor C is connected to the ignition switch and sensor D is connected to the headlights of the car. The alarm (X) connected to the system will be ON if the following conditions occur:

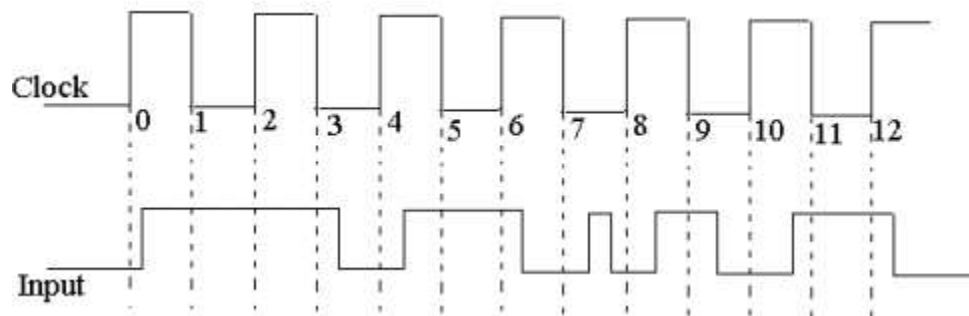
- If both doors are OPEN.
- If the headlights are ON while the ignition switch is OFF.
- If the door with sensor B is OPEN with headlights ON

- i. Determine the truth table for the system.
 - ii. Obtain the simplified:
 - a. SOP expression.
 - b. POS expression
 - iii. Implement the circuit using:
 - a. **NOR** gates only
 - b. A suitable Multiplexer.
- b) Explain the differences between a Combinational and a Sequential Logic Circuit.

(20 Marks)

Question 3

- i. The clock and the input waveform shown below are applied to the input of a positive edge triggered D flip-flop. Sketch the output waveforms. Assume the initial flip-flop value to



be zero.

- ii. Write the complete truth table of a **T Flip-flop** and derive its Excitation table.
- iii. Design a counter to count the sequence **2, 3 5, 7**. Use positive edge triggered **T** flip- flops and gates.
- iv. Three edge-triggered JK flip flops in a synchronous circuit have the following input conditions:

$$\begin{array}{ll}
 J_A = Q_C & K_A = 1 \\
 J_B = 1 & K_B = Q_A \cdot Q_C \\
 J_C = Q_B & K_C = 1
 \end{array}$$

Assume that the initial state is $Q_A = 0$ $Q_B = 1$ $Q_C = 1$. Find the count sequence.

(20 Marks)

Question 4

- i. Briefly explain how N-Type semi-conductors are formed using its crystal structure.
- ii. Draw the Volt-Ampere characteristics of a practical diode and mark I_s (Reverse Saturation Current), PIV (Peak Inverse Voltage) and V_b (Forward Biased Voltage)
- iii. Draw the circuit diagram of Half Wave Rectifier and explain the operation of it.
- iv. Draw the circuit diagram of Full Wave Rectifier and explain the operation of it.

(20 Marks)

Question 5

- i. Draw I-V characteristic curves of ON and OFF switches.
- ii. Explain the reversed biased mode and forward biased mode of the P-N junction.
- iii. Figure 1 depicts a circuit containing a Silicon diode. The external voltage source is 10V. The resistor (R) is 2k ohms. Calculate followings.

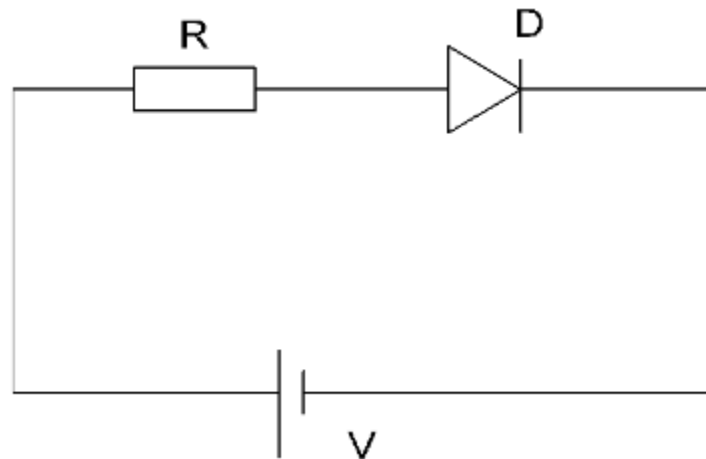


Figure 1

- a) Operating Mode of the Diode
- b) Potential Difference across the diode
- c) Potential Difference across the Resistor
- d) Current passing through the diode

(20 Marks)

Question 6

- What are the two types of Bipolar Junction Transistors? Draw their circuit symbols.
- Consider the following circuit and obtain the operating mode of the transistor. Hence Calculate I_B , I_C and V_{CE} . Assume that the transistor is made out of Silicon.

Material	$V_{CE}(\text{sat})$	$V_{BE}(\text{sat})$	$V_{BE}(\text{act})$	$V_{BE}(\text{cut in})$
Si	0.2V	0.8V	0.7V	0.5V
Ge	0.1V	0.3V	0.2V	0.1V

β	R_B	R_C	V_{CC}	V_{BB}
100	200k Ω	2k Ω	10V	5V

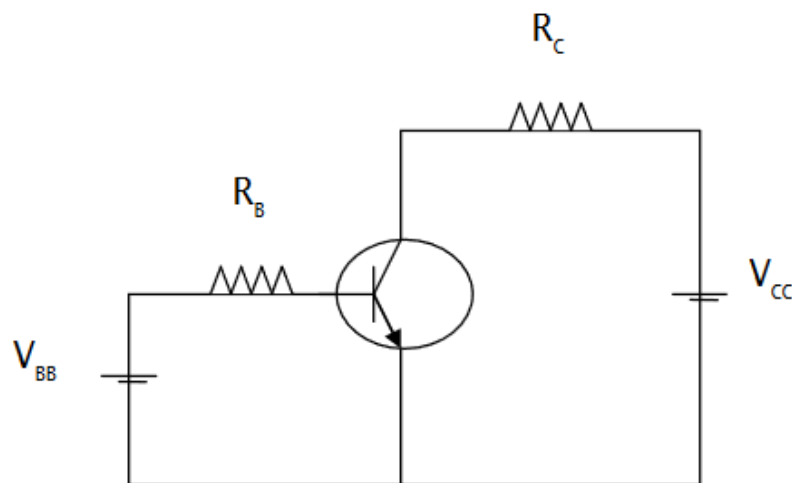


Figure 2

(20 Marks)